

# Supplementary information for “Graph-conditioned state-space models for genetic perturbation response prediction: a controlled evaluation against simple baselines”

Aryan Padarthi and Riley Huynh

## S1 Legacy perturbation encoding

Table S1 quantifies the encoding flaw described in Section 3.1 of the main text, computed with the original data loader (legacy/data\_loader.py) and split manifests (splits/splits\_manifest\_\*.json) by gcpm/audit\_hvg\_encoding.py. HVGs were selected on training cells only, as in the original pipeline.

Table S1: Legacy HVG-restricted encoding (Section 3.1): number of the 105 CRISPRa targets among the 500 HVGs, and perturbations whose input vector is all zero, for the ten split manifests of the earlier version of this work.

| Split | Targets in HVGs | Train all-zero | Distinct train inputs | Test all-zero | Test $n$ |
|-------|-----------------|----------------|-----------------------|---------------|----------|
| 0     | 17              | 133/169 (79%)  | 16                    | 52/66 (79%)   | 66       |
| 1     | 12              | 146/183 (80%)  | 14                    | 49/52 (94%)   | 52       |
| 2     | 14              | 147/174 (84%)  | 14                    | 43/61 (70%)   | 61       |
| 3     | 13              | 133/161 (83%)  | 14                    | 60/74 (81%)   | 74       |
| 4     | 13              | 134/167 (80%)  | 15                    | 61/68 (90%)   | 68       |
| 5     | 14              | 132/168 (79%)  | 17                    | 59/67 (88%)   | 67       |
| 6     | 14              | 133/161 (83%)  | 15                    | 58/74 (78%)   | 74       |
| 7     | 15              | 136/177 (77%)  | 19                    | 50/58 (86%)   | 58       |
| 8     | 14              | 137/167 (82%)  | 14                    | 58/68 (85%)   | 68       |
| 9     | 14              | 135/176 (77%)  | 19                    | 55/59 (93%)   | 59       |

## S2 Model development on the validation set of split 1

All design decisions were made on the validation perturbations of split 1; test perturbations were not used. Table S2 lists the successive changes and the lowest validation MSE (all genes) reached in short pilot runs (12–35 epochs). For reference, predicting no change gives 0.00433, the train mean 0.00361, the linear model 0.00312 and the additive baseline 0.00244 on the same perturbations (the latter using the training singles only).

Table S2: Pilot runs of GCP-Mamba on the validation perturbations of split 1.

| Change                                                                           | Best validation MSE | Observation                                               |
|----------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------|
| Initial model (scalar readout, raw PCA loadings)                                 | 0.00362             | equal to the train mean; training loss also at train mean |
| + bilinear readout                                                               | 0.00360             | unchanged                                                 |
| + standardised gene embeddings                                                   | 0.00569             | fits training data, generalises worse than the mean       |
| + per-target embeddings, dropout, weight decay $10^{-2}$                         | 0.00523             | large initial loss (random outputs on all genes)          |
| + initialisation at the train-mean prediction ( $\text{lr} = 2 \times 10^{-3}$ ) | 0.00320             | first improvement over the mean                           |
| + learning rate $10^{-3}$ (final)                                                | 0.00297             |                                                           |

### S3 Statistical comparisons and initialisation variance

Table S3 lists all pairwise tests between GCP-Mamba and the other models on the Norman data (deep models averaged over initialisations). Table S4 shows how much the split-level error of each model changes between initialisations; the within-split range is of the same size as the differences between graph-conditioned and graph-free variants. Table S5 gives the Pearson correlation on the top-20 DE genes for every model.

Table S3: GCP-Mamba versus every other model on the Norman data: two-sided Wilcoxon signed-rank tests on per-perturbation differences in MSE (top-20 DE genes), pooled over the five splits; Holm correction within each test group. Negative median differences favour GCP-Mamba.

| Test group       | Comparison model                                                   | $n$ | Median diff. | $P$     | $P_{\text{Holm}}$ |
|------------------|--------------------------------------------------------------------|-----|--------------|---------|-------------------|
| All test         | No change                                                          | 561 | -0.1704      | 1.1e-81 | 1.5e-80           |
| All test         | Train mean                                                         | 561 | -0.0523      | 7.6e-47 | 9.1e-46           |
| All test         | Additive                                                           | 561 | +0.0269      | 8.5e-23 | 8.5e-22           |
| All test         | Linear (PCA)                                                       | 561 | -0.0222      | 3.2e-12 | 2.2e-11           |
| All test         | Linear (ctrl emb.)                                                 | 561 | -0.0416      | 8.3e-46 | 9.1e-45           |
| All test         | GCP-Mamba ( $\Delta$ only)                                         | 561 | +0.0054      | 8.1e-07 | 4e-06             |
| All test         | GCP-Mamba ( $\Delta$ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$ ) | 561 | +0.0046      | 3.5e-05 | 0.00011           |
| All test         | GCP-Mamba (permuted graph)                                         | 561 | +0.0013      | 5.3e-05 | 0.00011           |
| All test         | GCP-Mamba (random order)                                           | 561 | +0.0064      | 1.5e-08 | 8.8e-08           |
| All test         | Mamba (no graph)                                                   | 561 | +0.0024      | 4.1e-06 | 1.6e-05           |
| All test         | Graph-MLP (no scan)                                                | 561 | -0.0002      | 0.83    | 0.83              |
| All test         | GCP-Mamba (additive prior)                                         | 561 | +0.0273      | 5.2e-22 | 4.7e-21           |
| All test         | Mamba (additive prior)                                             | 561 | +0.0281      | 1.6e-21 | 1.3e-20           |
| Unseen single    | No change                                                          | 185 | -0.0540      | 1.7e-19 | 2.2e-18           |
| Unseen single    | Train mean                                                         | 185 | -0.0023      | 0.06    | 0.42              |
| Unseen single    | Additive                                                           | 185 | -0.0043      | 0.00087 | 0.0087            |
| Unseen single    | Linear (PCA)                                                       | 185 | -0.0010      | 0.46    | 1                 |
| Unseen single    | Linear (ctrl emb.)                                                 | 185 | -0.0025      | 0.014   | 0.11              |
| Unseen single    | GCP-Mamba ( $\Delta$ only)                                         | 185 | +0.0006      | 0.44    | 1                 |
| Unseen single    | GCP-Mamba ( $\Delta$ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$ ) | 185 | -0.0003      | 0.35    | 1                 |
| Unseen single    | GCP-Mamba (permuted graph)                                         | 185 | -0.0010      | 0.0088  | 0.079             |
| Unseen single    | GCP-Mamba (random order)                                           | 185 | +0.0014      | 0.15    | 0.74              |
| Unseen single    | Mamba (no graph)                                                   | 185 | -0.0000      | 0.28    | 1                 |
| Unseen single    | Graph-MLP (no scan)                                                | 185 | -0.0008      | 0.11    | 0.66              |
| Unseen single    | GCP-Mamba (additive prior)                                         | 185 | -0.0049      | 2.6e-05 | 0.00032           |
| Unseen single    | Mamba (additive prior)                                             | 185 | -0.0056      | 6.7e-05 | 0.00073           |
| Double, 0/2 seen | No change                                                          | 32  | -0.2160      | 3.3e-08 | 4.2e-07           |
| Double, 0/2 seen | Train mean                                                         | 32  | -0.0487      | 0.0034  | 0.041             |
| Double, 0/2 seen | Additive                                                           | 32  | +0.0183      | 0.33    | 1                 |
| Double, 0/2 seen | Linear (PCA)                                                       | 32  | -0.0417      | 0.018   | 0.18              |
| Double, 0/2 seen | Linear (ctrl emb.)                                                 | 32  | -0.0464      | 0.005   | 0.055             |
| Double, 0/2 seen | GCP-Mamba ( $\Delta$ only)                                         | 32  | +0.0084      | 0.3     | 1                 |
| Double, 0/2 seen | GCP-Mamba ( $\Delta$ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$ ) | 32  | +0.0038      | 0.41    | 1                 |
| Double, 0/2 seen | GCP-Mamba (permuted graph)                                         | 32  | +0.0053      | 0.23    | 1                 |
| Double, 0/2 seen | GCP-Mamba (random order)                                           | 32  | +0.0058      | 0.61    | 1                 |
| Double, 0/2 seen | Mamba (no graph)                                                   | 32  | -0.0019      | 0.75    | 1                 |
| Double, 0/2 seen | Graph-MLP (no scan)                                                | 32  | +0.0006      | 0.83    | 1                 |
| Double, 0/2 seen | GCP-Mamba (additive prior)                                         | 32  | +0.0172      | 0.32    | 1                 |
| Double, 0/2 seen | Mamba (additive prior)                                             | 32  | +0.0067      | 0.59    | 1                 |
| Double, 1/2 seen | No change                                                          | 254 | -0.2765      | 1.8e-40 | 2.3e-39           |

Table S3: GCP-Mamba versus every other model on the Norman data: two-sided Wilcoxon signed-rank tests on per-perturbation differences in MSE (top-20 DE genes), pooled over the five splits; Holm correction within each test group. Negative median differences favour GCP-Mamba.

| Test group       | Comparison model                                                   | $n$ | Median diff. | $P$     | $P_{\text{Holm}}$ |
|------------------|--------------------------------------------------------------------|-----|--------------|---------|-------------------|
| Double, 1/2 seen | Train mean                                                         | 254 | -0.0995      | 3.9e-29 | 4.2e-28           |
| Double, 1/2 seen | Additive                                                           | 254 | +0.0575      | 2.2e-21 | 1.9e-20           |
| Double, 1/2 seen | Linear (PCA)                                                       | 254 | -0.0389      | 3.5e-08 | 2.4e-07           |
| Double, 1/2 seen | Linear (ctrl emb.)                                                 | 254 | -0.0800      | 8.8e-30 | 1.1e-28           |
| Double, 1/2 seen | GCP-Mamba ( $\Delta$ only)                                         | 254 | +0.0106      | 0.00029 | 0.00086           |
| Double, 1/2 seen | GCP-Mamba ( $\Delta$ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$ ) | 254 | +0.0086      | 0.00071 | 0.0014            |
| Double, 1/2 seen | GCP-Mamba (permuted graph)                                         | 254 | +0.0046      | 2e-05   | 7.9e-05           |
| Double, 1/2 seen | GCP-Mamba (random order)                                           | 254 | +0.0127      | 4e-06   | 2.4e-05           |
| Double, 1/2 seen | Mamba (no graph)                                                   | 254 | +0.0064      | 5.5e-06 | 2.7e-05           |
| Double, 1/2 seen | Graph-MLP (no scan)                                                | 254 | +0.0020      | 0.03    | 0.03              |
| Double, 1/2 seen | GCP-Mamba (additive prior)                                         | 254 | +0.0576      | 8.2e-22 | 8.2e-21           |
| Double, 1/2 seen | Mamba (additive prior)                                             | 254 | +0.0610      | 8.1e-21 | 6.5e-20           |
| Double, 2/2 seen | No change                                                          | 90  | -0.3615      | 1.7e-16 | 2.3e-15           |
| Double, 2/2 seen | Train mean                                                         | 90  | -0.2286      | 1.7e-16 | 2.3e-15           |
| Double, 2/2 seen | Additive                                                           | 90  | +0.1065      | 6.4e-15 | 6.4e-14           |
| Double, 2/2 seen | Linear (PCA)                                                       | 90  | -0.0720      | 2e-05   | 7.9e-05           |
| Double, 2/2 seen | Linear (ctrl emb.)                                                 | 90  | -0.1615      | 4.2e-15 | 4.6e-14           |
| Double, 2/2 seen | GCP-Mamba ( $\Delta$ only)                                         | 90  | +0.0157      | 9.1e-05 | 0.00027           |
| Double, 2/2 seen | GCP-Mamba ( $\Delta$ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$ ) | 90  | +0.0239      | 7e-06   | 4.2e-05           |
| Double, 2/2 seen | GCP-Mamba (permuted graph)                                         | 90  | +0.0064      | 1.3e-05 | 6.3e-05           |
| Double, 2/2 seen | GCP-Mamba (random order)                                           | 90  | +0.0150      | 0.00035 | 0.0007            |
| Double, 2/2 seen | Mamba (no graph)                                                   | 90  | +0.0175      | 5.6e-06 | 3.9e-05           |
| Double, 2/2 seen | Graph-MLP (no scan)                                                | 90  | -0.0032      | 0.1     | 0.1               |
| Double, 2/2 seen | GCP-Mamba (additive prior)                                         | 90  | +0.1045      | 7e-15   | 6.4e-14           |
| Double, 2/2 seen | Mamba (additive prior)                                             | 90  | +0.1063      | 7.7e-15 | 6.4e-14           |

Table S4: Variation due to random initialisation on the Norman data. For each split, the MSE (top-20 DE genes, all test perturbations) of three independently initialised runs; shown are the mean over splits of the within-split range and standard deviation, and the mean split-level MSE.

| Model                      | Mean MSE | Within-split s.d. | Within-split range |
|----------------------------|----------|-------------------|--------------------|
| GCP-Mamba                  | 0.290    | 0.024             | 0.046              |
| GCP-Mamba (permuted graph) | 0.285    | 0.023             | 0.042              |
| Mamba (no graph)           | 0.278    | 0.016             | 0.032              |
| Graph-MLP (no scan)        | 0.288    | 0.017             | 0.033              |
| GCP-Mamba (additive prior) | 0.225    | 0.001             | 0.003              |

## S4 Hyperparameters and compute

Table S5: Pearson correlation of predicted and observed expression change on the top-20 DE genes (mean  $\pm$  s.d. over five splits; higher is better)

| Model                                                              | Unseen single   | Double, 0/2 seen | Double, 1/2 seen | Double, 2/2 seen | All test        |
|--------------------------------------------------------------------|-----------------|------------------|------------------|------------------|-----------------|
| No change                                                          | —               | —                | —                | —                | —               |
| Train mean                                                         | $0.53 \pm 0.05$ | $0.72 \pm 0.12$  | $0.66 \pm 0.06$  | $0.67 \pm 0.07$  | $0.62 \pm 0.04$ |
| Additive                                                           | $0.54 \pm 0.05$ | $0.70 \pm 0.15$  | $0.82 \pm 0.03$  | $0.97 \pm 0.01$  | $0.75 \pm 0.03$ |
| Linear (PCA)                                                       | $0.52 \pm 0.07$ | $0.64 \pm 0.18$  | $0.73 \pm 0.03$  | $0.84 \pm 0.04$  | $0.67 \pm 0.03$ |
| Linear (ctrl emb.)                                                 | $0.52 \pm 0.05$ | $0.70 \pm 0.09$  | $0.69 \pm 0.06$  | $0.76 \pm 0.08$  | $0.65 \pm 0.05$ |
| GCP-Mamba                                                          | $0.55 \pm 0.08$ | $0.71 \pm 0.13$  | $0.76 \pm 0.03$  | $0.86 \pm 0.04$  | $0.71 \pm 0.04$ |
| GCP-Mamba ( $\Delta$ only)                                         | $0.57 \pm 0.06$ | $0.71 \pm 0.15$  | $0.76 \pm 0.02$  | $0.87 \pm 0.03$  | $0.72 \pm 0.03$ |
| GCP-Mamba ( $\Delta$ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$ ) | $0.56 \pm 0.07$ | $0.72 \pm 0.14$  | $0.76 \pm 0.03$  | $0.88 \pm 0.03$  | $0.71 \pm 0.04$ |
| GCP-Mamba (permuted graph)                                         | $0.53 \pm 0.08$ | $0.68 \pm 0.16$  | $0.76 \pm 0.03$  | $0.86 \pm 0.03$  | $0.70 \pm 0.04$ |
| GCP-Mamba (random order)                                           | $0.58 \pm 0.08$ | $0.73 \pm 0.12$  | $0.77 \pm 0.03$  | $0.88 \pm 0.02$  | $0.72 \pm 0.04$ |
| Mamba (no graph)                                                   | $0.55 \pm 0.08$ | $0.70 \pm 0.14$  | $0.77 \pm 0.03$  | $0.87 \pm 0.03$  | $0.71 \pm 0.04$ |
| Graph-MLP (no scan)                                                | $0.55 \pm 0.06$ | $0.68 \pm 0.17$  | $0.77 \pm 0.02$  | $0.86 \pm 0.03$  | $0.70 \pm 0.03$ |
| GCP-Mamba (additive prior)                                         | $0.53 \pm 0.06$ | $0.70 \pm 0.17$  | $0.82 \pm 0.03$  | $0.98 \pm 0.01$  | $0.74 \pm 0.03$ |
| Mamba (additive prior)                                             | $0.54 \pm 0.06$ | $0.70 \pm 0.18$  | $0.82 \pm 0.03$  | $0.98 \pm 0.01$  | $0.75 \pm 0.04$ |

Table S6: Hyperparameters shared by all deep-model variants.

| Parameter                                | Value                                                            |
|------------------------------------------|------------------------------------------------------------------|
| Layers, model width $d$ , state size $N$ | 2, 32, 4                                                         |
| Readout rank                             | 16                                                               |
| Gene embedding                           | 64 PCs of control-cell expression, standardised                  |
| Co-expression graph                      | $ r $ on control cells, $k = 20$ neighbours, symmetrised         |
| Graph signal                             | 3 diffusion steps, decay 0.5, $\log(1 + 10^3 \cdot)$             |
| Scan chunk length, log-decay clamp       | 16, $-5$                                                         |
| Optimiser                                | AdamW, lr $10^{-3}$ , weight decay $10^{-2}$ , batch 8, clip 1.0 |
| Dropout (perturbation encoder)           | 0.1                                                              |
| Epochs                                   | min. 40, max. 100, patience 15 on validation MSE                 |
| Seeds                                    | model seed = split seed (1–5)                                    |
| Hardware                                 | 4-core CPU, no GPU                                               |